Martingale Innovations from Recurrent Networks for Sequential Change-Point Detection

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Sequential monitoring is difficult when observations are serially dependent, high-dimensional, and lack a trusted parametric model. This paper revisits the innovations principle with a learned predictor: a recurrent network removes the serially predictable component of the stream, and a CUSUM statistic monitors the resulting prediction innovations. Its aim is foundational: the false-alarm and delay guarantees of classical change-point theory carry over, essentially intact, to a black-box learned predictor. The development is two-tier. First, for contractive recurrent networks, the hidden state admits a Doob decomposition into a predictable part and a martingale-difference innovation that stabilizes geometrically and whose partial sums obey a functional central limit theorem. Second, for any predictor with an almost-sure bound on the conditional residual drift and sub-Gaussian tails, the Innovation CUSUM has an exponential lower bound on the in-control average run length—with the best exponent available uniformly over that class—and a logarithmic upper bound on the detection delay. In the Gaussian mean-shift case the statistic coincides with the classical likelihood-ratio CUSUM and is first-order minimax optimal. The guarantee needs no parametric model for the observed dynamics, though it is not assumption-free; we also give conditions under which a learning guarantee verifies the drift condition with high probability. We also identify when learning helps: under a nonlinear conditional mean, linear residualization has an irreducible defect that a learned predictor removes. Simulations and real-data examples illustrate the theory.

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1. Introduction

Streams of data that must be monitored for structural change are now routine: intensive-care units track dozens of physiological signals, public-health agencies monitor infectious-disease incidence, industrial systems log hundreds of correlated sensor channels, and risk desks follow panels of asset returns. In each case, a detector must decide online whether the mechanism generating the stream has changed. The two possible errors have different costs. A *false alarm* — firing while the system remains in control — may trigger an unnecessary clinical intervention, halt a production line, or prompt a premature portfolio rebalancing. A *missed detection*, or a long detection delay, may leave a deteriorating patient unmonitored, an equipment fault undetected, or a loss unhedged. The statistical problem is therefore not merely to detect change, but to control false alarms without paying for that control through excessive delay.

This trade-off is especially difficult in the settings that motivate this paper. The observations are serially dependent; the dependence structure is typically unknown in advance and rarely specified with enough confidence to support a parametric monitor; and the number of monitored coordinates may be large. Sequential change-point detection under these three conditions — unknown serial dependence, limited model specification, and growing dimension — is the problem studied here.

Several commonly used tools solve parts of this problem. Classical CUSUM procedures are powerful — indeed optimal — under independence or a correctly specified parametric model (Page, 1954, Lorden, 1971), while offline segmentation (Killick, Fearnhead and Eckley, 2012) and Bayesian online methods

(Adams and MacKay, 2007) offer flexible alternatives. These approaches, however, generally lose their false-alarm guarantees once the serial dependence structure is unknown or the observation dimension grows. Classical time-series models handle dependence explicitly: autoregressive models fix the lag order, hidden Markov models encode memory through a finite state space, and Kalman filters propagate a Gaussian sufficient statistic (Box et al., 2015, Hamilton, 1994). They inherit clean guarantees when the model is correct, but only at the price of requiring the model to be correct. What is missing is a monitoring scheme whose false-alarm behavior remains valid under unknown serial dependence and growing dimension.

A further strand of literature monitors *residuals* rather than raw observations. Monitoring with estimated parameters goes back at least to Chu, Stinchcombe and White (1996); the asymptotics of sequential residual empirical processes were developed by Bai (1994); Lai (1995) treats sequential detection in dynamical systems; and Aue and Horváth (2013) survey structural-break detection for time series. These procedures monitor the residuals of a *specified* parametric class, and their guarantees are tied to that class being correct. In a complementary direction, conformal test martingales and e-detectors (Vovk, Gammerman and Shafer, 2005, Shin, Ramdas and Rinaldo, 2024) provide anytime-valid nonparametric monitoring under exchangeability-type conditions or a known in-control reference. The present paper sits between the two strands: like the residual tradition it monitors one-step prediction errors, but the predictor is learned rather than specified; like the e-detector tradition it avoids a parametric model of the dynamics, but the in-control requirement is an explicit residual drift-and-tail condition rather than exchangeability, and the monitor is the classical Page CUSUM.

This paper takes a route with a long pedigree in time-series analysis: transform the observations into innovations, then monitor the innovations. The innovations principle — that the one-step prediction error of an optimal predictor carries no serially predictable component — underlies Wiener–Kolmogorov prediction theory, the innovations approach to filtering (Kailath, 1968), residual diagnostics in Box–Jenkins modeling (Box et al., 2015), and residual-based detection of abrupt changes (Basseville and Nikiforov, 1993). In the linear-Gaussian world the principle is exact: Kalman-filter innovations are independent, and a CUSUM applied to them inherits the classical false-alarm guarantees. Outside that world, however, the filter must be specified correctly — precisely what is unavailable when the dependence structure is unknown. Recurrent neural networks (RNNs) (Rumelhart, Hinton and Williams, 1986), long short-term memory networks (LSTMs) (Hochreiter and Schmidhuber, 1997), gated recurrent units (GRUs) (Cho et al., 2014), and selective state-space models such as Mamba (Gu and Dao, 2023) learn predictive state representations from data rather than fixing the memory structure in advance. They therefore supply a natural modern implementation of the innovations principle: use a learned predictor to remove the serially predictable component, and monitor what remains. The question this paper answers is what survives of the classical CUSUM guarantees when the filter is learned rather than specified.

Two questions must be answered. First, in what precise probabilistic sense does a learned recurrent state generate innovations, and how quickly do those innovations stabilize, so that calibration performed on a training segment remains valid at monitoring time? Second, what sequential-monitoring guarantee does a CUSUM applied to learned innovations inherit, and does that guarantee withstand growing observation dimension?

Our answers are deliberately two-tier. At the first tier, for the basic RNN recurrence under an operator-norm contraction condition on the recurrent weight matrix (Assumption 2.2) — a condition the practitioner can *enforce* through spectral normalization (Miyato et al., 2018) rather than merely hope for — the hidden state admits a classical Doob decomposition into a predictable component and a martingale-difference innovation. The innovation converges geometrically to its stationary counterpart, and the accumulated innovations obey a functional central limit theorem.

At the second tier, the detector's guarantee is architecture-agnostic by design. Exact Bayes prediction residuals are martingale differences (Proposition 2.7); fitted residuals are useful to the extent that their conditional mean drift is small. The contractive-RNN result supports this condition through a bridge bound (Proposition 2.8), which separates three sources of error: transient hidden-state initialization, readout approximation to the Bayes predictor, and estimation error. The CUSUM theory itself then assumes directly that the fitted residuals have bounded conditional drift and conditional sub-Gaussian tails. Under those operating conditions it yields an in-control average-run-length lower bound (Theorem 3.2) and an out-of-control detection-delay upper bound (Theorem 3.6), which together give the exponential-false-alarm / logarithmic-delay trade-off (Corollary 3.7). Thus the RNN stabilization theorem motivates and partially verifies the residual condition, while the CUSUM guarantee applies to any learned predictor satisfying that condition.

The contributions organize around four connected claims. First, under operator-norm contractivity, recurrent hidden states generate stabilized martingale-difference innovations: the hidden state is uniformly L^2-bounded, decomposes into a predictable component plus an innovation, and the innovation converges geometrically to its stationary counterpart (Theorem 2.3). The accumulated innovations then satisfy a martingale functional central limit theorem (Corollary 2.6). Second, a new bridge proposition translates this state-level stabilization into a residual-level approximation bound for recurrent predictors with Lipschitz readouts (Proposition 2.8). Third, for fitted prediction residuals satisfying explicit conditional-drift and tail assumptions, the Innovation CUSUM has a sharp exponential in-control average-run-length lower bound and a corresponding out-of-control detection-delay upper bound (Theorems 3.2 and 3.6); together they attain the minimax-optimal detection rate of classical sequential analysis, exactly so under a Gaussian innovation shift (Theorem 3.9). Fourth, we identify *when* learning helps: under a nonlinear conditional mean, any fixed linear residualization carries an irreducible innovation defect, whereas a consistent learned predictor drives this defect to zero (Proposition 2.9), improving the guaranteed in-control ARL through the drift margin (Remark 2.10). The empirical studies then examine these mechanisms: learned innovations are closer to martingale differences under misspecification, calibration is more stable when the in-control window is short, and equal-ARL comparisons delimit when the learned detector is competitive with correctly specified parametric procedures.

The contribution lies in this synthesis. The individual ingredients are classical — the Doob decomposition of an adapted square-integrable process; the martingale-difference property of Bayes prediction residuals; the geometric forgetting of initial conditions under a contractive stochastic recursion, standard in iterated-random-function and mixing theory (Doukhan, 1994) and used elsewhere for stable recurrent models (Miller and Hardt, 2019); the martingale functional central limit theorem (Hall and Heyde, 1980, Billingsley, 1999); the L^2-orthogonal-projection characterization of the best linear predictor (Proposition 2.9); and CUSUM theory dating to Page (1954) and Lorden (1971), including the exponential-supermartingale argument used for its false-alarm guarantee. None of these building blocks is new on its own. What appears to be new is their combination: using contractive recurrent states to construct stabilized innovation processes, connecting these innovations to fitted prediction residuals, and deriving a nonparametric sequential-monitoring guarantee for the resulting approximate-MDS CUSUM under explicit residual-drift and tail conditions.

This perspective is distinct from most existing theory for recurrent architectures, which emphasizes stability or expressivity (Miller and Hardt, 2019, Miyato et al., 2018, Gonon and Ortega, 2020), and from neural tangent kernel and information-bottleneck analyses, which primarily concern feedforward networks or fixed-time representations (Jacot, Gabriel and Hongler, 2018, Tishby and Zaslavsky, 2015, Saxe et al., 2019). Structured state-space models such as Mamba (Gu, Goel and Ré, 2022, Gu and Dao, 2023) have likewise been studied mainly from the standpoint of efficient sequence modeling (Orvieto et al., 2023). Here the focus is different: the learned state is used to remove the serially predictable component of the stream, so that the remaining innovation process can be monitored with model-free sequential guarantees under explicit drift, tail, and post-change signal assumptions.

The theoretical framework and main results appear in Sections 2–3, followed by four empirical studies in Section 4. Eight additional studies, background definitions, and all formal proofs are given in the

Supplementary Material. Table B.1 summarizes the assumptions and conclusions of the main results and makes explicit the two-tier structure of the paper: contractive-RNN innovation theory in Section 2 and architecture-agnostic detection theory in Section 3.

2. Methodology

We work on a complete probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $(X_t)_{t\geq 1}$ be a strictly stationary ergodic process; by Kolmogorov's extension theorem we may, and do, extend it to a two-sided strictly stationary ergodic process $(X_t)_{t\in\mathbb{Z}}$. Throughout Sections 2–3 the filtration is the *two-sided* natural filtration

$$\mathcal{F}_t \;=\; \sigma(X_s : s \leq t) \vee \sigma(h_0), \qquad t \in \mathbb{Z},$$

where the RNN initial state h_0 (introduced below) is square-integrable and independent of $(X_t)_{t\in\mathbb{Z}}$; a deterministic h_0, as in all our experiments, is a special case. Conditioning on the two-sided past is what makes the stationary innovation process of Section 2.1 exactly stationary and adapted; the detector of Section 3 conditions only on the observable one-sided history, and its guarantees are self-contained there. The tail σ-algebra is $\mathcal{T}_\infty = \bigcap_{t\geq 1} \sigma(X_t, X_{t+1}, \ldots)$. Throughout Sections 2–3 we use p to denote the hidden-state dimension and d_{in} for the input dimension. An RNN processes the sequence via $h_t = \phi(W_h h_{t-1} + W_x x_t + b_h)$ (with $h_t \in \mathbb{R}^p$) and an LSTM augments this with a gated cell state; both architectures are defined precisely in Section A of the Supplementary Material. In Section 4 the symbol d refers to the observation (input) dimension of the multivariate DGP, following the change-point literature convention. Throughout, we assume the effective recurrent Lipschitz constant satisfies $L_\phi \|W_h\|_{\text{op}} < 1$ (operator-norm contractivity; see Assumption 2.2 below), which can be encouraged in practice by spectral normalization (Miyato et al., 2018). For one-step prediction of a target Y_{t+1}, the central object is the innovation $e_{t+1} = Y_{t+1} - \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$, whose martingale-difference sequence (MDS) property is stated in Proposition 2.7. A process (X_t) is *absolutely regular* (β-mixing) if its mixing coefficients $\beta(k) = \sup_t \mathbb{E}\{\sup_{B\in\sigma(X_{t+k}, X_{t+k+1}, \ldots)} |\mathbb{P}(B \mid \mathcal{F}_t) - \mathbb{P}(B)|\}$ tend to zero. Such assumptions are used only for empirical motivation and local approximation, not to claim that (h_t) alone is a Markov chain. Full definitions, the reverse martingale convergence theorem, and the i.i.d. Hewitt–Savage zero-one law are collected in Section A of the Supplementary Material.

Remark 2.1 (Non-tail-measurable targets). Parts (d) and (e) show that when the target θ is not tail-measurable, the forward and reverse limits *differ*: the forward limit $\mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$ may incorporate information from the finite initial segment, while the reverse limit $\mathbb{E}[\theta \mid \mathcal{T}_\infty]$ discards it. This formalizes *selective forgetting*: the reverse martingale extracts only the structurally persistent (tail-measurable) component of any target.

Assumption 2.2 (Effective operator-norm contractivity). Throughout the paper we assume $L_\phi \|W_h\|_{\text{op}} < 1$, where L_ϕ is the Lipschitz constant of ϕ and $\|W_h\|_{\text{op}} = \sigma_{\max}(W_h)$. Spectral normalization (Miyato et al., 2018) can encourage this condition in practice.

2.1. Martingale Decomposition of RNN Hidden States

Throughout this subsection we work under Assumption 2.2 ($\rho := L_\phi \|W_h\|_{\text{op}} < 1$). The existence and uniqueness of the stationary causal solution $(H_t^*)_{t\in\mathbb{Z}}$ satisfying $H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h)$, $H_t^* \in \sigma(X_t, X_{t-1}, \ldots)$, and the geometric coupling bound $\left\|h_t - H_t^*\right\|_{L^2} \leq \rho^t \left\|h_0 - H_0^*\right\|_{L^2}$ are proved directly in Proposition B.3 (Section B.16 of the Supplementary Material), so this paper's theorem chain does not depend on any external or unpublished result. The companion paper Chang (2026) studies the complementary question of hidden-state convergence under the same contraction condition from a reverse-filtration perspective; none of its results are used here. We write $\Delta_0 := \left\|h_0 - H_0^*\right\|_{L^2}$, $D_t^* := \mathbb{E}[H_t^* \mid \mathcal{F}_{t-1}]$ (stationary predictable component), and $M_t^* := H_t^* - D_t^*$ (stationary MDS innovation). Here the two-sided filtration of the setup is essential: H_t^* is a fixed measurable functional of $(X_s)_{s\le t}$, hence $\mathcal{F}_t$-measurable, and D_t^* is the same functional of the shifted path at every t (conditioning on the independent $\sigma(h_0)$ leaves it unchanged), so $(M_t^*)_{t\in\mathbb{Z}}$ is a strictly stationary, ergodic MDS *adapted* to $(\mathcal{F}_t)$. With the one-sided filtration $\sigma(X_1,\dots,X_{t-1})$ neither exact stationarity nor adaptedness would hold, because the pre-sample past influences H_t^*. The stationary causal solution is generally a *random* process; it should not be identified with a deterministic mean-field fixed point or a tail projection.

The Doob decomposition in part (ii) of the following theorem is, on its own, elementary: subtracting its conditional expectation turns any square-integrable adapted process into a predictable term plus a martingale difference. The substance of Theorem 2.3 is what the contractive recursion adds to this decomposition — uniform L^2 control, the *geometric stabilization* of the innovation toward its stationary counterpart (iv), and the stationary innovation covariance Σ_M^* (v) — which are the ingredients the functional CLT (Corollary 2.6) and the Innovation CUSUM (Section 3) actually consume.

Theorem 2.3 (MDS Decomposition with Stabilization Rate). *Let $(X_t)_{t\ge1}$ be a strictly stationary ergodic process with $\mathbb{E}\|X_t\|^2 < \infty$, and let $(h_t)_{t\ge0}$ be the RNN hidden-state sequence with initial state h_0 as in the setup (square-integrable, independent of the process). Set $\rho = L_\phi\|W_h\|_{\mathrm{op}} < 1$ and $\Delta_0 = \left\|h_0 - H_0^*\right\|_{L^2}$. Then, for each $t \ge 2$:*

(i) ***(L^2-boundedness.)*** *(h_t) is uniformly L^2-bounded. When $\phi(0) = 0$ (e.g.* tanh*, ReLU):*

$$\sup_t \|h_t\|_{L^2} \le \|h_0\|_{L^2} + \frac{L_\phi\left(\|W_x\|_{\mathrm{op}}\|X_1\|_{L^2} + \|b_h\|\right)}{1-\rho} =: C_0 < \infty.$$

For activations with $\phi(0) \ne 0$ (e.g. sigmoid), an additional term $\|\phi(0)\|/(1-\rho)$ is added to C_0 (from $\|\phi(u)\| \le \|\phi(0)\| + L_\phi\|u\|$, so the constant enters undamped by L_ϕ); finiteness is unchanged.

(ii) ***(MDS decomposition.)*** *The Doob decomposition $h_t = D_t + M_t$ holds, where $D_t := \mathbb{E}[h_t \mid \mathcal{F}_{t-1}]$ is $\mathcal{F}_{t-1}$-measurable (predictable) and $M_t := h_t - D_t$ satisfies $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$ a.s. The pair $(M_t, \mathcal{F}_t)_{t\ge2}$ is a martingale difference sequence.*

(iii) ***(Innovation L^2 bound.)*** *Writing* $\mathrm{Var}_2(X_t) := \mathbb{E}\|X_t - \mathbb{E}X_t\|^2$,

$$\mathbb{E}\|M_t\|^2 \le L_\phi^2\|W_x\|_{\mathrm{op}}^2\,\mathrm{Var}_2(X_t) =: \sigma_M^2. \tag{1}$$

(iv) ***(Innovation stabilization rate.)*** *The MDS innovations converge to their stationary counterparts at a geometric rate:*

$$\left\|M_t - M_t^*\right\|_{L^2} \le 2\rho^t\Delta_0. \tag{2}$$

(v) ***(Conditional variance decomposition and stationary innovation covariance.)*** *The law-of-total-variance decomposition holds exactly:*

$$\mathbb{E}\|h_t - \mathbb{E}h_t\|^2 = \mathbb{E}\|D_t - \mathbb{E}h_t\|^2 + \mathbb{E}\|M_t\|^2. \tag{3}$$

In the stationary regime the innovation covariance matrix

$$\Sigma_M^* := \mathbb{E}\left[M_t^*(M_t^*)^\top\right] \tag{4}$$

is well-defined, independent of t, positive semi-definite, and satisfies $\Sigma_M^* \preceq \sigma_M^2 I_p$. Moreover $\lambda_{\min}(\Sigma_M^*) > 0$ under the following nondegeneracy condition: for every nonzero $v \in \mathbb{R}^p$,

$$\mathrm{Var}\big(v^\top \phi(W_h H_{t-1}^* + W_x X_t + b_h) \,\big|\, \mathcal{F}_{t-1}\big) > 0 \quad a.s. \tag{5}$$

A sufficient condition for (5) is that X_t is not entirely predictable from $\mathcal{F}_{t-1}$ and W_x has full row rank rank$(W_x) = p$ *under linear activation; this sufficient condition need not apply as stated for nonlinear ϕ, and can fail for saturating nonlinearities or when $d_{\mathrm{in}} < p$ (see Remark 2.5).*

Proof. See Section B of the Supplementary Material. □

Remark 2.4. Parts (i)–(iii) parallel the earlier Proposition 2.7 argument but now apply to the hidden state itself rather than the prediction target. **Parts (iv) and (v) are new.** Part (iv) shows that the transient MDS innovation approaches the stationary innovation at the same geometric rate ρ^t as the hidden state itself, which justifies asymptotic calibration of the Innovation CUSUM after a burn-in of order $\log(\Delta_0/\varepsilon)/(-\log\rho) \approx \log(\Delta_0/\varepsilon)/(1-\rho)$ steps to reach tolerance ε. Part (v) provides the stationary innovation covariance Σ_M^* needed for the functional central limit theorem (CLT) and for Mahalanobis scoring in the multivariate CUSUM.

Remark 2.5 (Scope of the nondegeneracy condition). The nondegeneracy condition (5) is the cleanest sufficient condition for $\lambda_{\min}(\Sigma_M^*) > 0$: it requires the innovation M_t^* to have non-zero variance in every direction. Full row rank rank$(W_x) = p$ is a natural structural sufficient condition under linear activation, but need not imply (5) for saturating nonlinearities (tanh, sigmoid) when the conditional support of X_t lies where the activation is nearly flat. In the common regime $d_{\mathrm{in}} < p$ (e.g. scalar input, $p = 32$ hidden units) W_x cannot have full row rank, so that linear sufficient condition is unavailable, and under a linear or locally linearized activation Σ_M^* is at most rank-d_{in}. That bound is specific to the linear case: a low-dimensional input transformed nonlinearly can yield rank larger than d_{in} (a scalar Z mapped through coordinates responding to Z and to Z^2 already populates two directions). Whether Σ_M^* is rank-deficient when $d_{\mathrm{in}} < p$ therefore depends on the activation and cannot be read off from rank(W_x) alone; in practice one should verify positive definiteness from a pilot run, and the functional CLT (Corollary 2.6) assumes it rather than deriving it from W_x.

Corollary 2.6 (Functional CLT for MDS innovations). *In addition to the hypotheses of Theorem 2.3, assume $\mathbb{E}\|X_t\|^{2+\delta} < \infty$ for some $\delta > 0$ and that Σ_M^* is positive definite. Define the partial-sum process $\mathbf{S}_T(u) := T^{-1/2}\sum_{t=1}^{\lfloor Tu\rfloor} M_t$, $u \in [0,1]$. Then*

$$\mathbf{S}_T(\cdot) \xrightarrow{d} (\Sigma_M^*)^{1/2}\, W(\cdot) \qquad \text{in } D\big([0,1],\mathbb{R}^p\big), \tag{6}$$

where W is a standard p-dimensional Brownian motion and $D([0,1],\mathbb{R}^p)$ denotes the Skorokhod space of right-continuous left-limited functions (the natural domain for the step-function partial-sum process). The transient-correction process $\mathbf{T}_T(u) := T^{-1/2}\sum_{t=1}^{\lfloor Tu\rfloor}(M_t - M_t^*)$ satisfies $\sup_{u\in[0,1]}\|\mathbf{T}_T(u)\| \to 0$ in probability, with explicit rate

$$\mathbb{E}\left[\sup_{u\in[0,1]}\|\mathbf{T}_T(u)\|\right] \le \frac{2\Delta_0\rho}{\sqrt{T}(1-\rho)} \to 0$$

(see proof for the sup-norm bound and the functional Slutsky argument), so (6) holds for (M_t) and (M_t^*) simultaneously.

Proof. See Section B of the Supplementary Material. □

Proposition 2.7 (Bayes prediction residuals are martingale differences). *Let Y_{t+1} be a square-integrable, $\mathcal{F}_{t+1}$-measurable prediction target and let $m_t = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$. Then*

$$e_{t+1} = Y_{t+1} - m_t$$

is $\mathcal{F}_{t+1}$-measurable, satisfies $\mathbb{E}[e_{t+1} \mid \mathcal{F}_t] = 0$, and hence is a martingale difference relative to $(\mathcal{F}_t)$. If an $\mathcal{F}_t$-measurable estimator $\hat{m}_t$ obeys $\sup_t \|\hat{m}_t - m_t\|_{L^2} \le \eta$, then the fitted residuals $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ are approximate martingale differences in the sense that $\sup_t \|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2} \le \eta$.

Proof. See Section B of the Supplementary Material. □

The next proposition makes explicit how the contractive-RNN stabilization result enters the prediction-residual theory. It does not assert that the hidden state is itself the Bayes predictor. Rather, it separates the transient error caused by the initial recurrent state from the approximation and estimation errors of the readout used for prediction.

Proposition 2.8 (Bridge from recurrent states to prediction innovations). *Work under the hypotheses of Theorem 2.3. Let $Y_{t+1} \in L^2$ be a prediction target with Bayes predictor $m_t = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$. Let $o : \mathbb{R}^p \to \mathbb{R}^q$ be a readout with Lipschitz constant L_o, and define the recurrent predictor $\hat{m}_t = o(h_t)$. With H_t^* denoting the stationary causal hidden state and $\Delta_0 = \left\|h_0 - H_0^*\right\|_{L^2}$,*

$$\|\hat{m}_t - m_t\|_{L^2} \le L_o\, \rho^t \Delta_0 + \left\|o(H_t^*) - m_t\right\|_{L^2}. \tag{7}$$

Consequently the fitted prediction residual $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ satisfies

$$\|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2} \le L_o\, \rho^t \Delta_0 + \left\|o(H_t^*) - m_t\right\|_{L^2}. \tag{8}$$

If the readout belongs to an estimated class and its stationary readout error obeys $\left\|o(H_t^*) - m_t\right\|_{L^2} \le a_T + \xi_T$, where a_T is approximation error and ξ_T is estimation error, then the residual drift is bounded by $L_o \rho^t \Delta_0 + a_T + \xi_T$.

Proof. See Section B of the Supplementary Material. □

Proposition 2.7 identifies the innovation defect $\|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2}$ with the predictor's L^2 bias $\|m_t - \hat{m}_t\|_{L^2}$. The next result uses this identity to explain *why a learned predictor is preferable to a fixed parametric residualization* when the conditional mean is nonlinear: a linear (e.g. autoregressive) predictor carries an irreducible innovation defect equal to the nonlinearity of m_t, whereas a consistent learned predictor drives the defect to zero.

Proposition 2.9 (Innovation defect and the misspecification gap). *Let (X_t) be strictly stationary with $m_t = \mathbb{E}[X_{t+1} \mid \mathcal{F}_t] \in L^2$, and let $\hat{m}_t$ be any $\mathcal{F}_t$-measurable predictor with fitted residual $\hat{e}_{t+1} = X_{t+1} - \hat{m}_t$.*

(i) ***(Defect = bias.)*** *$\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t] = m_t - \hat{m}_t$, so the innovation defect is $\|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2} = \|m_t - \hat{m}_t\|_{L^2}$, which vanishes iff $\hat{m}_t = m_t$ a.s.*

(ii) ***(Irreducible linear defect.)*** *Let $L_t \subseteq L^2(\Omega, \mathcal{F}_t, \mathbb{P})$ be the closed subspace of affine functions of the observed history and $m_t^{\text{lin}} = \Pi_{L_t} X_{t+1}$ the best linear predictor. By stationarity the linear-residual defect is the constant*

$$\eta_{\text{lin}} = \operatorname{dist}_{L^2}(m_0, L_0) \ge 0,$$

and $\eta_{\text{lin}} > 0$ whenever $m_0 \notin L_0$, i.e. the process is nonlinear in mean.

(iii) ***(Consistent learning removes it.)*** *If $\|\hat{m}_t - m_t\|_{L^2} \le \eta_T$ with $\eta_T \to 0$, then the innovation defect $\to 0$; the learned residuals are asymptotically an exact MDS irrespective of the nonlinearity of m_0.*

Proof. See Section B of the Supplementary Material. □

Remark 2.10 (Consequence for calibration and the ARL guarantee). In the change-point theory of Section 3 the reference value must exceed the innovation defect, $k > \eta$, and the ARL guarantee improves with the net margin $\kappa = k - \eta$ (Theorem 3.2). Under nonlinearity in mean, a linear residualization is capped at $\kappa \le k - \eta_{\text{lin}}$ with $\eta_{\text{lin}} > 0$ fixed by the process, whereas a consistent learned predictor attains $\kappa \to k$. A learned predictor therefore yields a strictly better guaranteed in-control ARL than any fixed linear filter when the conditional mean is nonlinear; Study CUSUM-MISSPEC (Supplementary Material) exhibits exactly this gap, with the learned innovations markedly whiter than linear-whitening residuals.

Proposition 2.9 is qualitative: a consistent learned predictor removes the defect. The next result makes the removal *quantitative*, decomposing the defect exactly into an estimation and an approximation part. It is a *conditional* rate calculation: the rates in part (ii) are assumed, not established here for the recurrent class.

Proposition 2.11 (Rate of the innovation defect; conditional calculation). *Let $\hat{m} = g_{\hat{\phi}}$ be the recurrent predictor fitted by empirical risk minimization over a class Φ from T in-control observations, and write $\eta_T = \sup_t \|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2}$ for its innovation defect, $\sigma_e^2 = \mathbb{E}\|X_{t+1} - m_t\|^2$ for the irreducible one-step Bayes risk, and $R(\phi) = \mathbb{E}\|X_{t+1} - g_\phi(h_t)\|^2$ for the population risk (in the stationary regime, so constant in t).*

(i) ***(Exact bias–variance split.)*** *The squared defect equals the excess prediction risk and splits into estimation and approximation parts,*

$$\eta_T^2 = R(\hat{\phi}) - \sigma_e^2 = \underbrace{\Big(R(\hat{\phi}) - \inf_{\phi\in\Phi} R(\phi)\Big)}_{\varepsilon_{\text{est}}} + \underbrace{\Big(\inf_{\phi\in\Phi} R(\phi) - \sigma_e^2\Big)}_{\varepsilon_{\text{app}}},$$

where $\varepsilon_{\text{app}} = \inf_{\phi\in\Phi} \mathbb{E}\|m_t - g_\phi(h_t)\|^2$ is the class approximation error for the Bayes predictor.

(ii) ***(Rate, under assumed approximation and estimation bounds.)*** *Suppose the width-W class Φ_W approximates the Bayes predictor at rate $\varepsilon_{\text{app}}(\Phi_W) \le c_{\text{A}} W^{-2s}$ with smoothness index $s > 0$ (the form available for smooth filters approximated by recurrent/reservoir maps, Gonon and Ortega, 2020), and obeys, for β-mixing in-control data, an excess-risk bound $\mathbb{E}\,\varepsilon_{\text{est}} \le c_{\text{E}} W \log T / T$ (Mohri and Rostamizadeh, 2010). Balancing at $W \asymp (T/\log T)^{1/(2s+1)}$ gives $\mathbb{E}\,\eta_T = O\big((T/\log T)^{-s/(2s+1)}\big) = \widetilde{O}\big(T^{-s/(2s+1)}\big)$. Both bounds are* hypotheses*: the cited results do not establish these exact rates for the recurrent class used here, whose parameter count grows faster than W and for which mixing enters through an effective sample size.*

(iii) ***(What this does* not *give.)*** *The defect η_T is an L^2 quantity, whereas Assumption 3.1(B1) requires an* almost-sure *bound on $\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]$. An L^2 rate does not imply an essential-supremum bound, so η_T may* not *be substituted for η in Theorem 3.2: part (ii) controls a diagnostic, not the constant entering the ARL guarantee. Proposition 2.12 supplies the additional uniform control under which a learning guarantee does verify (B1).*

The next proposition closes that gap, making explicit the price: control must be *uniform* over the reachable state space, not merely in L^2.

Proposition 2.12 (High-probability verification of the drift condition). *Work under Assumption 2.2 and suppose:*

- (H1) ***(Bounded inputs.)*** *$\|X_t\| \le B_X$ a.s.; then $\|h_t\| \le C_h$ a.s. for an explicit constant C_h (see the proof), so h_t and H_t^* lie in the compact set $\mathcal{H} := \{h : \|h\| \le C_h\}$.*
- (H2) ***(Uniform approximation.)*** *The readout class contains an L_o-Lipschitz $o_\star$ with $a_\infty := \operatorname{ess\,sup}_t |o_\star(H_t^*) - m_t| < \infty$.*
- (H3) ***(Uniform estimation, with high probability.)*** *The fitted readout $\hat{o}$ satisfies, with probability at least $1 - \delta_T$ over the training sample $\mathcal{D}_T$, $\sup_{h\in\mathcal{H}} |\hat{o}(h) - o_\star(h)| \le \varepsilon_T$.*

Then, on that training event, $\left|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\right| \le a_\infty + \varepsilon_T + L_o\rho^t\Delta_0^\infty$ a.s. for every t, where $\Delta_0^\infty := \operatorname{ess\,sup}\left\|h_0 - H_0^*\right\|$. Consequently, for $t \ge t_0$ with $L_o\rho^{t_0}\Delta_0^\infty \le \varepsilon_T$, Assumption 3.1(B1) holds with $\bar{\eta}_T := a_\infty + 2\varepsilon_T$; and if $k > \bar{\eta}_T$ then, conditionally on $\mathcal{D}_T$, Theorem 3.2 gives $\mathbb{E}_0[\tau_h \mid \mathcal{D}_T] \ge \exp\{2(k - \bar{\eta}_T)h/\sigma^2\}$ on an event of probability at least $1 - \delta_T$.

Proof. See Section B of the Supplementary Material. □

Remark 2.13 (What (H3) costs). (H1)–(H2) are structural; (H3) is the substantive price, demanding sup-norm rather than L^2 estimation error. On the compact $\mathcal{H}$ this is the natural regime—uniform convergence over a bounded Lipschitz readout class follows from covering-number arguments under β-mixing—but we do not establish (H3) for a specific architecture, and regard doing so as the main route to making η genuinely estimable.

Proof. See Section B of the Supplementary Material. □

2.2. Extension to Non-Stationary Processes

The results in Section 2.1 require stationarity. We now describe how they extend to the locally stationary setting of Dahlhaus (1997).

Definition 2.14 (Locally stationary process; Dahlhaus 1997). A triangular array $(X_{t,T})_{1\le t\le T}$ is *locally stationary* with transfer function $A^0(u,\cdot)$ if there exists a representation

$$X_{t,T} = \int_{-\pi}^{\pi} e^{i\lambda t} A^0(t/T, \lambda)\, dZ(\lambda),$$

where $Z(\lambda)$ is a complex-valued orthogonal increment process and $A^0(u,\lambda)$ is a smooth function of the rescaled time $u = t/T \in [0,1]$.

Corollary 2.15 (Approximate local stationary decomposition). *Under the locally stationary model of Definition 2.14, assume that $A^0(u,\lambda)$ is Lipschitz in u uniformly in λ (Dahlhaus, 1997, Assumption 2), and that the array admits an L^2 stationary approximation satisfying $\sup_{t\in B_T(u)} \|X_{t,T} - X_t^{(u)}\|_{L^2} = O(b_T + m_T/T) + o(1)$ (which follows from the Lipschitz condition via Dahlhaus 1997). Fix $u \in (0,1)$ and let $X_t^{(u)}$ denote the stationary approximating process with transfer function $A^0(u,\cdot)$. Let $h_{t,T}$ and $h_t^{(u)}$ be the corresponding RNN states driven by $X_{t,T}$ and $X_t^{(u)}$, respectively, under the same contractive* *recurrence and a common square-integrable initialization at the left edge of the local block. Let $B_T(u)$ be any block of at most m_T consecutive time indices satisfying $|t/T-u| \le b_T$ for all $t \in B_T(u)$. If $b_T \downarrow 0$, $m_T \to \infty$, and $m_T/T \to 0$, then*

$$\sup_{t\in B_T(u)} \left\| h_{t,T} - h_t^{(u)} \right\|_{L^2} = O\left(b_T + \frac{m_T}{T}\right) + o(1). \tag{9}$$

The stationary approximation $h_t^{(u)}$ has the decomposition of Theorem 2.3, with all bounds evaluated under the local law at rescaled time u and spectral density $f(u,\lambda) = |A^0(u,\lambda)|^2$. Thus the martingale-difference decomposition is local and approximate rather than an almost-sure convergence statement to a literal "local tail" limit of the finite triangular array.

Proof. See Section B of the Supplementary Material. $\square$

3. Innovation CUSUM: In-Control False-Alarm Theory

We now give a rigorous in-control performance guarantee for the Innovation CUSUM, the change-point detector that applies a Page CUSUM to learned prediction innovations. This is the second tier of the theory: exact Bayes residuals form an MDS, fitted residuals are approximately MDS when their conditional drift is small, and this residual-level property drives the alarm-rate bound through a sub-Gaussian exponential supermartingale argument. No architecture-specific contraction assumption enters the CUSUM theorem; the guarantee applies to any predictor whose residuals satisfy the conditional drift and tail conditions below. Throughout this section, $(\mathcal{F}_t)$ denotes the observable (one-sided) filtration generated by the monitored data and any training information to which the fitted residuals $\hat{e}_t$ are adapted; the two-sided convention of Section 2 is not used here.

3.1. Setup and Assumptions

The one-sided Page CUSUM (Page, 1954) is

$$S_0 = 0, \qquad S_t = (S_{t-1} + \hat{e}_t - k)^+, \qquad \tau_h = \inf\{t \ge 1 : S_t \ge h\}, \tag{10}$$

where $k > 0$ is a reference (slack) value and $h > 0$ is the alarm threshold.

Assumption 3.1 (Innovation CUSUM operating conditions).

(B1) **(Conditional residual-drift bound.)** In the in-control regime the fitted prediction innovations satisfy

$$\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \le \eta \quad \text{a.s. for all } t,$$

for some finite constant $\eta \ge 0$. When $\hat{e}_t = Y_t - \hat{m}_{t-1}$ and $m_{t-1} = \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}]$, this condition is equivalent to the uniform bound $m_{t-1} - \hat{m}_{t-1} \le \eta$ a.s.; the L^2 bounds in Propositions 2.7 and 2.8 should be read as diagnostics or sufficient ingredients under additional uniform-control assumptions, not as a replacement for this conditional drift requirement.

(B2) **(Conditional sub-Gaussianity.)** In the in-control regime there exists $\sigma > 0$ such that $\mathbb{E}[\exp\{\lambda(\hat{e}_t - \mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}])\} \mid \mathcal{F}_{t-1}] \le e^{\lambda^2\sigma^2/2}$ for all $\lambda \in \mathbb{R}$. This holds when $\hat{e}_t \in [-B, B]$ a.s. with $\sigma = B$.

(B3) **(Reference value exceeds conditional drift.)** $k > \eta$; write $\kappa := k - \eta > 0$ for the *net drift margin*.

3.2. Main Result: Exponential ARL Lower Bound

The in-control average run length $\mathrm{ARL}_0 = \mathbb{E}_0[\tau_h]$ is the false-positive metric: a larger ARL_0 means rarer false alarms. Theorem 3.2 below bounds it from below; the complementary false-negative metric — the out-of-control detection delay — is bounded from above in Theorem 3.6 of Section 3.3, and the two combine into the classical exponential-ARL / logarithmic-delay trade-off (Corollary 3.7).

Theorem 3.2 (Sharp in-control ARL lower bound for the Innovation CUSUM). *Under Assumption 3.1, the in-control average run length (ARL_0; here $\mathbb{E}_0[\cdot]$ denotes expectation under the in-control distribution, i.e. prior to any change-point) satisfies*

$$\mathbb{E}_0[\tau_h] \;\geq\; \exp\left(\frac{2\kappa h}{\sigma^2}\right), \tag{11}$$

where $\kappa = k - \eta$ is the net drift margin and σ^2 the sub-Gaussianity parameter. The exponent $\theta^* = 2\kappa/\sigma^2$ is the sub-Gaussian adjustment exponent *of the increment $\hat{e}_t - k$: it is the largest tilt certified by (B1)–(B2), and it coincides with the exact Lundberg coefficient—and hence with the classical i.i.d. CUSUM rate—in the Gaussian benchmark. Equivalently, an in-control ARL of at least L is guaranteed by $h \geq (\sigma^2/2\kappa)\ln L$.*

Proof. See Section B of the Supplementary Material. The bound is obtained by a renewal (excursion) argument: at each return of the CUSUM to zero, the conditional probability that the ensuing excursion reaches h is at most $e^{-\theta^* h}$ by optional stopping on the exponential supermartingale $\exp\{\theta^* \sum(\hat{e}_j - k)\}$, so the number of excursions before an alarm stochastically dominates a geometric variable with mean $e^{\theta^* h}$. No union bound is used. □

Remark 3.3 (Sharpness). The ARL grows exponentially in h at the rate $2\kappa/\sigma^2$, where $\kappa = k - \eta$ is the net margin between the reference value and the in-control conditional residual drift; the bound degrades gracefully as η increases, becoming vacuous only when $\eta \to k$. The exponent is best possible *uniformly over the class* defined by Assumption 3.1: the Gaussian mean-shift increment attains it, so no exponent exceeding $2\kappa/\sigma^2$ is valid for every law satisfying (B1)–(B2), and there it matches the exact exponential rate of the classical i.i.d. CUSUM (Siegmund, 1985, Lorden, 1971). For a particular non-Gaussian increment law the exact Lundberg root may be larger, so the bound is sharp for the class rather than for every member of it. In contrast to an elementary Chernoff-plus-union-bound estimate, the pre-exponential constant is unity. The threshold formula $h \geq (\sigma^2/2\kappa)\ln L$ enables principled calibration from η and σ^2 alone, without distributional knowledge.

Remark 3.4 (Calibration in practice). The operating constants of Assumption 3.1 are estimated on a held-out in-control segment, disjoint from both the training data and the monitored stream. The theorem requires an upper bound on the *conditional* residual drift $\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]$, not merely a small unconditional mean; in practice this is assessed by conservative diagnostics—the calibration residual mean, autocorrelation tests, and, when available, residual-on-past regressions or the nonlinear predictability checks of Section 4.2. These give an empirical proxy for η; they do *not* constitute a theorem-valid estimator of the essential supremum in Assumption 3.1(B1). The sub-Gaussian proxy σ is the calibration standard deviation of $\hat{e}_t$ (or $\sigma = B$ when $\hat{e}_t \in [-B, B]$, as in (B2)); k is set with a comfortable margin $\kappa = k - \eta > 0$, and h follows from Theorem 3.2 for the target ARL L. Because (11) depends on η and σ only through upper bounds, over-estimating either is conservative: it lowers the guaranteed ARL rather than invalidating the bound. Section 4 reports the sensitivity of ARL_0 and delay to (k, h), separating theoretical from finite-sample diagnostic calibration.

Corollary 3.5 (Two-sided Innovation CUSUM). *Let $\tau_h^{\pm} = \min(\tau_h^+, \tau_h^-)$ be the alarm time of the two-sided CUSUM, where τ_h^+ is the one-sided alarm time of (10) and τ_h^- that of the mirrored recursion $S_t^- = (S_{t-1}^- - \hat{e}_t - k)^+$. If Assumption 3.1 holds for both $\hat{e}_t$ and $-\hat{e}_t$ (in particular, $|\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]| \le \eta$ a.s. in-control), then $\mathbb{E}_0[\tau_h^{\pm}] \ge \frac{1}{4}\exp(2\kappa h/\sigma^2)$, with the same exponent $2\kappa/\sigma^2$ as in Theorem 3.2. The empirical studies in Section 4 use this two-sided version.*

Proof. See Section B of the Supplementary Material. A bound of the form $\mathbb{E}_0[\tau_h^{\pm}] \ge \mathbb{E}_0[\tau_h]/2$ does *not* follow from the one-sided result, since the expectation of a minimum of two stopping times is not controlled by the average of their individual expectations. Instead the per-excursion bound of Theorem 3.2 gives, for each arm, the tail estimate $\mathbb{P}_0(\tau_h^{\pm} \le n) \le 2n\, e^{-\theta^* h}$ by a union bound over the two arms, and summing $\mathbb{P}_0(\tau_h^{\pm} > n)$ over $n \le \lfloor \frac{1}{2} e^{\theta^* h} \rfloor$ yields the stated bound. □

3.3. Out-of-Control Detection Delay

Theorem 3.2 controls the false-positive side. We now bound the false-negative side — the detection delay after a change — completing the sequential-monitoring guarantee. The delay is controlled once the change induces a positive drift in the innovation, the out-of-control counterpart of the in-control conditional-drift condition (B1).

Theorem 3.6 (Out-of-control detection-delay upper bound). *Consider the one-sided Innovation CUSUM (10) with a change-point at time τ. Suppose:*

(O1) ***(Post-change drift.)*** *There is $\mu > 0$ with $\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \ge k + \mu$ a.s. for all $t > \tau$;*

(O2) ***(Bounded increments.)*** *$|\hat{e}_t - k| \le B$ a.s.*

Then the expected detection delay, conditional on no false alarm before the change, satisfies

$$\mathbb{E}\big[\, \tau_h - \tau \;\big|\; \tau_h > \tau \,\big] \;\le\; \frac{h + B}{\mu}. \tag{12}$$

Proof. See Section B of the Supplementary Material. □

Corollary 3.7 (Exponential-ARL / logarithmic-delay trade-off). *Under Assumptions 3.1 and (O1)–(O2), choosing the threshold $h = (\sigma^2/2\kappa)\ln L$ gives simultaneously an in-control guarantee $\mathbb{E}_0[\tau_h] \ge L$ and a detection-delay guarantee $\mathbb{E}[\tau_h - \tau \mid \tau_h > \tau] \le \mu^{-1}\big((\sigma^2/2\kappa)\ln L + B\big) = O(\ln L/\mu)$.*

Proof. Combine the threshold formula of Theorem 3.2 with (12). □

Remark 3.8 (A model-free guarantee under residual conditions). Corollary 3.7 has the classical sequential-analysis form: false-alarm protection grows exponentially in the threshold, while detection delay grows only logarithmically in the target ARL. The result is model-free with respect to the data-generating dynamics, but it is not distribution-free in the assumption-free sense. It requires the in-control residual-drift and tail conditions of Assumption 3.1 and the post-change signal and boundedness conditions (O1)–(O2). Under those conditions, no parametric model for the observed process is needed. That the resulting $O(\log L)$ delay is in fact order-optimal — and *exactly* minimax optimal in the Gaussian case — is made precise in Theorem 3.9 below.

3.4. Asymptotic Optimality

Corollary 3.7 shows the Innovation CUSUM attains the optimal *order* of detection delay. We now record that it is order-optimal within Lorden's minimax framework, and is *exactly* minimax optimal in the canonical Gaussian case. For a stopping time τ and change-point $\nu \geq 1$, Lorden's worst-case detection delay and the false-alarm constraint class are

$$\mathcal{J}(\tau) = \sup_{\nu \geq 1} \operatorname{ess\,sup} \mathbb{E}_\nu\big[(\tau - \nu + 1)^+ \,\big|\, \mathcal{F}_{\nu-1}\big], \qquad C_\gamma = \{\tau : \mathbb{E}_\infty[\tau] \geq \gamma\},$$

where $\mathbb{E}_\nu$ (resp. $\mathbb{E}_\infty$) denotes the law with change at ν (resp. no change).

Theorem 3.9 (Asymptotic optimality of the Innovation CUSUM). *Set $h = h_\gamma = (\sigma^2/2\kappa)\log\gamma$, so that $\tau_{h_\gamma} \in C_\gamma$ by Theorem 3.2.*

(i) ***(Order optimality.)*** *Under Assumption 3.1 and (O1)–(O2), $\mathcal{J}(\tau_{h_\gamma}) \leq (\sigma^2/2\kappa\mu)\log\gamma + B/\mu = O(\log\gamma)$, so the delay scales logarithmically in γ, with* detection efficiency $2\kappa\mu/\sigma^2$. *If in addition the change admits a Kullback–Leibler information number, i.e. the conditional log-likelihood ratios satisfy $n^{-1}\sum_{t=\nu}^{\nu+n-1}\log\{dP_\nu(\cdot \mid \mathcal{F}_{t-1})/dP_\infty(\cdot \mid \mathcal{F}_{t-1})\} \to I \in (0,\infty)$ under the regularity of Lorden (1971), then this matches* in order *the universal lower bound $\inf_{\tau \in C_\gamma}\mathcal{J}(\tau) \geq (\log\gamma)/I\,(1 - o(1))$, so the detector is order-optimal; first-order optimality additionally requires the efficiency to equal I, which is the case of part (ii). Assumption 3.1 and (O1)–(O2) alone constrain residual moments, not likelihood ratios, and so do not by themselves define I.*

(ii) ***(Gaussian innovations: efficiency attains I, and exact optimality under exact calibration.)*** *Suppose that, conditionally on $\mathcal{F}_{t-1}$, the innovation $\hat{e}_t$ is Gaussian with variance σ^2 and mean η in control and $\eta + \delta$ after the change ($\delta > 0$), and that the reference is the midpoint $k = \eta + \delta/2$ (so $\kappa = \mu = \delta/2$). Then the detection efficiency equals the information number, $2\kappa\mu/\sigma^2 = \delta^2/(2\sigma^2) = I$, and the Innovation CUSUM coincides with the Page likelihood-ratio CUSUM for the change; hence $\mathcal{J}(\tau_{h_\gamma}) \sim (\log\gamma)/I$, i.e. τ_{h_γ} is* first-order *asymptotically minimax optimal. Exact Lorden optimality (Moustakides, 1986) holds for the Page CUSUM whose threshold is calibrated so that its in-control ARL equals γ* exactly*; the analytic threshold h_γ guarantees only $\mathbb{E}_\infty[\tau_{h_\gamma}] \geq \gamma$ and is in general conservative, so exact optimality is not claimed for h_γ itself.*

Proof. See Section B of the Supplementary Material. □

Remark 3.10 (Interpretation). Part (i) compresses the operating characteristic into a single scalar, the detection efficiency $2\kappa\mu/\sigma^2$: the product of the in-control margin $\kappa = k - \eta$ and the post-change margin μ, scaled by the innovation variance. Part (ii) shows this efficiency is not merely of the optimal order but attains the information number I exactly when the change acts through a Gaussian shift of the innovation mean — the setting in which the innovation is the efficient score. The midpoint reference $k = \eta + \delta/2$ that the classical CUSUM design prescribes is precisely the choice maximizing $\kappa\mu$ subject to $\kappa + \mu = \delta$.

3.5. Dimension Stability

Corollary 3.11 (Dimension-stable ARL). *Let $\hat{e}_t \in \mathbb{R}^d$ be the multivariate LSTM innovation and define the aggregated Mahalanobis score*

$$\mathcal{Z}_t := \frac{1}{d}(\hat{e}_t - \mu_0)^\top \Sigma_0^{-1}(\hat{e}_t - \mu_0) - 1,$$

where $\mu_0 = \mathbb{E}[\hat{e}_t]$ and $\Sigma_0 = \mathrm{Var}(\hat{e}_t)$. Suppose Assumption 3.1 holds for the innovation vector with $\eta_d = O(1)$ in the aggregate sense $\|\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]\| \le \eta_d$ a.s. This is a high-dimensional residual-drift assumption on the full vector, not merely a coordinatewise accuracy condition. Take a fixed reference value $k_d \equiv k > 0$, write $u_t := \hat{e}_t - \mu_0$ and $b_t := \mathbb{E}[u_t \mid \mathcal{F}_{t-1}]$ for its conditional mean (so $\|b_t\|_2 \le 2\eta_d$ a.s. by the drift assumption above), and assume that the conditionally centered *deviation satisfies*

$$\mathbb{E}\left[(u_t - b_t)(u_t - b_t)^\top \mid \mathcal{F}_{t-1}\right] \preceq \Sigma_0 \quad a.s.$$

(conditional homoskedasticity of the centered innovation). This bounds only the conditional variance *of $\hat{e}_t$, not the squared drift $b_t b_t^\top$, and is therefore strictly weaker than assuming $\mathbb{E}[u_t u_t^\top \mid \mathcal{F}_{t-1}] \preceq \Sigma_0$ directly; the latter would force $\mathbb{E}[\mathcal{Z}_t \mid \mathcal{F}_{t-1}] \le 0$ with no residual term, which is not the regime intended here. Assume further that the in-control innovation covariance is uniformly non-degenerate: $\lambda_{\min}(\Sigma_0) \ge c_0 > 0$ for all d, so that $\|\Sigma_0^{-1}\|_{\mathrm{op}} = O(1)$ as $d \to \infty$. Assume also that the scalar score $\mathcal{Z}_t$ satisfies Assumption 3.1(B2) with sub-Gaussianity parameter $\sigma_{Z,d} = O(1)$ as $d \to \infty$; this holds, for example, when the innovations $\hat{e}_t$ are uniformly bounded in d. Then $\eta'_d := 4\|\Sigma_0^{-1}\|_{\mathrm{op}}\eta_d^2/d = O(d^{-1}) \to 0$, the net drift margin $\kappa_d = k_d - \eta'_d \to k > 0$ as $d \to \infty$, and for all sufficiently large d the ARL lower bound (11) holds with $\kappa_d \ge k/2 > 0$.*

By contrast, for a raw vector autoregression (VAR(1)) process $X_t = \Phi X_{t-1} + \varepsilon_t$ with coefficient matrix $\Phi \in \mathbb{R}^{d\times d}$, $\|\Phi\|_{\mathrm{op}} = \rho_{\mathrm{op}} > 0$, $\varepsilon_t \overset{\mathrm{iid}}{\sim} (0, I_d)$, $\mu_0 = 0$, stationary covariance $\Sigma_X := \mathrm{Var}(X_t)$, and $\Sigma_0 = I_d$ (identity standardization), the aggregated Mahalanobis score satisfies

$$\mathbb{E}[\mathcal{Z}_t^{\mathrm{raw}} \mid \mathcal{F}_{t-1}] = \frac{1}{d}\,\|\Phi X_{t-1}\|^2 = \frac{1}{d}\,\mathrm{tr}\!\left(\Phi X_{t-1} X_{t-1}^\top \Phi^\top\right), \tag{13}$$

(the noise-variance term $d^{-1}\mathrm{tr}(I_d) - 1 = 0$ cancels exactly when $\Sigma_0 = I_d$), and this conditional bias need not vanish as $d \to \infty$: for the symmetric case $\Phi = \rho_{\mathrm{op}} I_d$, by the law of large numbers applied to the i.i.d. stationary components of ΦX_{t-1}, $d^{-1}\|\Phi X_{t-1}\|^2 \to \rho_{\mathrm{op}}^2/(1-\rho_{\mathrm{op}}^2) > 0$ a.s.; for general Φ the bias $d^{-1}\mathrm{tr}(\Phi\Sigma_X\Phi^\top)$ is strictly positive for each fixed d whenever $\Phi \ne 0$ and $\Sigma_X \succ 0$, but need not remain bounded away from zero as $d \to \infty$ (e.g. for $\Phi = \mathrm{diag}(\rho_{\mathrm{op}}, 0, \ldots, 0)$, the bias $d^{-1}\mathrm{tr}(\Phi\Sigma_X\Phi^\top) \to 0$). Consequently the net drift margin $\kappa_d^{\mathrm{raw}} = k - \eta_d^{\mathrm{raw}}$ cannot be made positive by a fixed reference value k unless k is recalibrated to exceed the serial-dependence bias η_d^{raw}, which itself depends on Φ and the stationary distribution of X_{t-1} in a way that raw CUSUM calibration does not capture.

Proof. See Section B of the Supplementary Material. □

Remark 3.12 (Connection to Study CUSUM-DIM). Corollary 3.11 provides a formal account of the dimension-scaling pattern observed in Study CUSUM-DIM (Section 4.4): at $d = 10$, the raw and whitened CUSUM ARL_0 falls to 47 and 42 while the Innovation CUSUM rises to 365. Equation (13) shows that the serial-dependence bias η_d^{raw} remains nonzero after the $1/d$ normalization for any $\rho_{\mathrm{op}} > 0$, so a fixed reference value k cannot provide a positive drift margin without explicit recalibration using Φ and Σ_X. This is a calibration requirement, not a universal impossibility: if the reference value k could be perfectly matched to the bias, the raw CUSUM would also be valid. The Innovation CUSUM guarantee holds for all sufficiently large d (specifically once $\eta'_d = 4\|\Sigma_0^{-1}\|_{\mathrm{op}}\eta_d^2/d < k$, which holds for $d \ge d_0$ where d_0 depends on η_d and k).

4. Empirical Studies and Discussion

This section presents four empirical studies, one for each step of the theory. A reminder on scope: the contractive-RNN theory of Section 2 supplies the innovation/stabilization prototype, whereas the detector evaluated below is LSTM-based and rests on the approximate Bayes-residual MDS condition of Assumption 3.1, not on an LSTM contraction theorem. The studies therefore probe that condition directly — through residual-whiteness diagnostics — before turning to detection performance. Study MDS examines the MDS property (Theorem 2.3) in controlled stationary settings. Study CUSUM-CP then compares a CUSUM built on LSTM innovations with classical detectors at a known change-point and fixed threshold, reporting false alarms and detections separately and checking the in-control run length against Theorem 3.2; the matched-calibration version of this comparison is Study CUSUM-EQARL. Study CUSUM-DIM examines the dimension-stability of the Innovation CUSUM (Corollary 3.11) across $d \in \{1, 2, 5, 10\}$, finding results consistent with the theoretical prediction. Study ETF applies the full pipeline to a real multivariate dataset — US equity sector returns around the 2020 COVID crash.

Eight supporting studies are deferred to the Empirical Supplement (Section D of the Supplementary Material). Study MDS-VAR extends the MDS diagnostic to multivariate VAR(1) processes; Study KAL benchmarks the LSTM against the Kalman filter on clean and noisy state-space models; Study NILE applies the three-way CUSUM comparison to the Nile River annual discharge series (Cobb, 1978); and Study PELT contrasts the Innovation CUSUM with the PELT offline detector across the same dimensions as Study CUSUM-DIM. Study CUSUM-EQARL revisits the dimension-scaling comparison under a matched false-alarm level and a matched estimation budget, showing that the fixed-threshold advantage seen in Study CUSUM-DIM is a finite-sample calibration effect rather than a structural one; Study CUSUM-MISSPEC then exhibits the regime — a nonlinear, misspecified process — in which the learned innovations are genuinely whiter and detection is faster than linear whitening, as Proposition 2.9 predicts. Study LS-Diag examines locally stationary AR data and finds that a stationary-trained LSTM forget gate systematically mis-adapts to smooth non-stationarity, motivating an explicit non-stationarity correction; Study J-Surr benchmarks four data-driven surrogates for the predictive-decay profile J_t (defined in Section 4.1), of which only a Kullback–Leibler importance estimation procedure (KLIEP)-corrected estimator (Sugiyama, Suzuki and Kanamori, 2012) achieves normalized mean squared error (NMSE) below one under smooth variation, while all four deteriorate under abrupt change.

All experiments use small-to-moderate sample sizes deliberately, since this paper focuses on theoretical diagnostics rather than application performance. All code is in `code/` (Python 3.9, PyTorch 2.2, and R 4.1), reproducible via `python run_all.py`. Studies MDS and CUSUM-CP use 1000–2000 independent test sequences per configuration; Study CUSUM-DIM uses 1000 test and 2000 in-control sequences.

4.1. Processes, Benchmarks, and Real Data

Three synthetic data-generating processes and one real dataset are used in the main text; a second real series, the Nile River annual discharge, is analyzed in Study NILE of the Supplementary Material. The first synthetic process is the AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$, $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$, with $\phi \in \{0.3, 0.7, 0.95\}$. For the initialization transient starting at $h_0 = 0$, the transient predictive-decay profile is $J_t = |\phi|^t \cdot \sigma/\sqrt{1-\phi^2}$ (geometric decay).

The second is a two-state hidden Markov model (HMM) with states $s \in \{0, 1\}$, transition matrix $P = \left[\begin{smallmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{smallmatrix}\right]$, and emissions $X_t \mid s_t \sim \mathcal{N}(\pm 1.5, 1)$; the predictive-decay profile J_t is computed numerically via the forward algorithm. The third is a regime-switching autoregression (RS-AR) with AR coefficients $\phi_1 = 0.30$ and $\phi_2 = 0.85$ under the same transition matrix.

The real dataset is a panel of US equity sector exchange-traded fund (ETF) log-returns described in Section 4.5, examined in a univariate configuration ($d = 1$, using the Financial sector ETF XLF) and a three-dimensional configuration ($d = 3$, XLF together with the Technology sector ETF XLK and the Energy sector ETF XLE), with the 2020-02-19 COVID market peak as the known change-point.

4.2. Study MDS: Stationary Case — Prediction-Residual MDS Diagnostic

We train one `InstrumentedLSTM` (hidden dimension $p = 32$, spectral normalization on W_h, Adam, $N = 1000$ training sequences of length $T = 150$, 40 epochs) on each process and compute one-step-ahead residuals $r_t = X_{t+1} - \hat{y}_t$ on $N_{\text{test}} = 1000$ independent sequences of length $T = 200$. The Ljung–Box statistic (Ljung and Box, 1978) at maximum lag 5 is applied to each sequence; the *MDS pass rate* is the fraction of sequences not rejecting at level 0.05. This is a diagnostic for linear residual autocorrelation, not a proof of the martingale-difference property. For theorem-level calibration one would also examine conditional drift through residual-on-history regressions or nonlinear predictability checks; the Ljung–Box statistic is used here as a simple, reproducible first screen.

Table 1 shows the results. For AR(1) with $\phi \in \{0.3, 0.7\}$ and the HMM, the LSTM pass rate is close to the nominal level $1 - \alpha = 0.95$, consistent with Proposition 2.7 when the fitted predictor is close to the conditional mean. The lower rate for $\phi = 0.95$ (0.386) is expected: a near-unit-root process requires very long sequences to reach its stationary regime; within $T = 200$ steps the LSTM cell state has not fully converged, so the residuals retain residual autocorrelation. The OLS AR(1) baseline confirms that the pattern is process-driven, not a model artifact: OLS residuals for the HMM and RS-AR have notably low pass rates (0.561 and 0.947) because a misspecified linear predictor cannot capture the nonlinear regime structure.

Table 1. Study MDS: Ljung–Box MDS pass rate (fraction of 1000 test sequences not rejecting zero autocorrelation at $\alpha = 0.05$, lag ≤ 5). Bold values are within ± 0.05 of the nominal level 0.95.

Process	LSTM pass rate	OLS pass rate	Expected	N_{seq}
AR(1), $\phi = 0.30$	**0.948**	0.969	0.95	1000
AR(1), $\phi = 0.70$	0.905	0.974	0.95	1000
AR(1), $\phi = 0.95$	0.386	0.953	0.95	1000
HMM	**0.953**	0.561	0.95	1000
RS-AR	0.814	0.947	0.95	1000

4.3. Study CUSUM-CP: Innovation CUSUM at a Known Change-Point

Study MDS operated entirely under stationarity. A natural next question is whether an online CUSUM built on LSTM innovations can detect an abrupt structural break reliably, and whether it beats the two classical alternatives. This empirically examines the prediction of Theorem 3.2: if the LSTM residuals approximate a martingale difference sequence in-control, the ARL lower bound should be empirically visible.

For abrupt change-point AR, we generate 2000 sequences with a single abrupt change in the AR coefficient from $\phi_{\text{before}} = 0.30$ to $\phi_{\text{after}} = 0.85$ at $\tau^* = 100$ ($T = 200$). All detectors use Page's two-sided CUSUM (Page, 1954, Lorden, 1971) with the same threshold $h = 5$ and reference value $k = 0.5$, estimated from the first 20% of each sequence. Three methods are compared. M1 (Raw) applies CUSUM to the standardized observations $(X_t - \hat{\mu}_0)/\hat{\sigma}_0$, the standard i.i.d.-calibrated detector. M2 (Whitened) applies CUSUM to AR(1)-whitened residuals $(X_t - \hat{\phi} X_{t-1})/\hat{\sigma}_\varepsilon$, where $\hat{\phi}$ is estimated from the burn-in window; this is the best classical fix, as it explicitly removes lag-1 autocorrelation. M3 (the Innovation CUSUM) applies CUSUM to LSTM one-step-ahead prediction residuals, standardized by in-control moments estimated from a held-out stationary calibration set; *no knowledge of the AR structure is given to the method.*

The key issue with M1 is that the Page CUSUM was designed for i.i.d. input. For AR(ϕ) data, successive standardized observations $(X_t - \mu_0)/\sigma_0$ remain correlated with lag-1 autocorrelation $\approx \phi$, causing the CUSUM statistic to accumulate faster than expected. Concretely, the in-control average run length (ARL_0) — the expected number of steps before a false alarm — is only ≈ 119 for M1, versus the nominal i.i.d. value of 285 at $h = 5$. On sequences of length 200, this means M1 fires before τ^* in 60% of cases (Table 2).

Table 2 shows the three-way comparison at $h = 5$, reporting the two error types separately. The distinction matters here because the raw detector's apparent "detection" is an illusion of the first-passage rule: M1 issues an alarm on virtually every sequence, but 60.3% of those alarms fire *before* the change (false positives), leaving a correct-detection (true-positive) rate of only 39.7%. M2 (whitened, given the correct AR model) raises the correct-detection rate to 77.2%, and the Innovation CUSUM to 87.7% — the highest — while simultaneously holding the lowest false-positive rate (12.3% versus 22.4% and 60.3%). Misses (false negatives) are negligible for all three ($\leq 0.4\%$). The ARL_0 of M3 (≈ 436) matches that of M2 (≈ 440) and both exceed the i.i.d. nominal (285), indicating that in-control LSTM residuals are slightly under-dispersed relative to the i.i.d. calibration assumption, consistent with the theoretical ARL lower bound of Theorem 3.2 holding with slack.

Detection delay must be read with care, because the tabulated mean averages (alarm $- \tau^*$) over *all* alarms, false positives included. For M3, whose alarms are overwhelmingly correct, the +10.1-step mean is a genuine detection delay; for M1 the -16.3-step "delay" is not a delay at all but a restatement of its false-positive rate — its alarms predominantly precede the change. Read this way, the two error types tell a consistent story: the Innovation CUSUM attains both the lowest false-positive rate and the highest correct-detection rate, so its false-alarm control is not bought by missing changes. A conditional detection-delay analysis (delay given a correct post-τ^* detection), which removes the false-positive contamination entirely, is the natural complement to this table under a matched-ARL_0 calibration. A representative CUSUM trajectory on a single sequence, illustrating that the Innovation CUSUM alarms after the change without being given the AR structure, is shown in Figure D.1 of the Supplementary Material.

Separately, the LSTM's prediction mean squared error (MSE) more than doubles after the change-point (ratio $\mathrm{MSE}_{\mathrm{post}}/\mathrm{MSE}_{\mathrm{pre}} = 2.25$, versus 1.05 for an oracle model retrained on post-break data), quantifying the cost of model mis-specification after the break. Study J-Surr (Section D.9 of the Supplementary Material) compares four data-driven surrogates for the predictive-decay profile J_t (naive, forget-gate, block-bootstrap, and KLIEP-corrected Sugiyama, Suzuki and Kanamori 2012) and shows that the KLIEP-corrected estimator achieves NMSE below one under smooth non-stationarity, while all four fail under abrupt change.

4.4. Study CUSUM-DIM: Innovation CUSUM Dimension Scaling

The question Study CUSUM-CP leaves open is whether the Innovation CUSUM advantage documented in the scalar setting survives as the observation dimension d grows — the central claim of Corollary 3.11. Study CUSUM-DIM extends the three-way CUSUM comparison to VAR(1) processes with dimension $d \in \{1, 2, 5, 10\}$. The data-generating process is

$$X_t = \Phi X_{t-1} + \eta_t, \qquad \eta_t \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, I_d),$$

where $\Phi = U \operatorname{diag}(\lambda) U^\top$ is constructed so that $\|\Phi\|_{\mathrm{op}} = \max_i |\lambda_i|$. Here and below, ρ_{op} denotes the operator norm of the VAR coefficient matrix, distinct from the RNN contraction constant $\rho = L_\phi \|W_h\|_{\mathrm{op}}$ of Section 2. A change-point at $\tau = 100$ (within a length-$T = 200$ window) raises the operator norm

Table 2. Study CUSUM-CP: three-way CUSUM comparison at $h = 5$. AR(1) change $\phi : 0.30 \to 0.85$ at $\tau^* = 100$ ($T = 200$, $N = 2000$ sequences). Each replicate is classified by its first alarm into one of three mutually exclusive outcomes: a *false positive* (FP, alarm before τ^*), a *correct detection* (TP, first alarm at or after τ^*), or a *miss* (FN, no alarm in the window); FP + TP + FN = 1. Mean delay‡ is the average of (alarm $-\ \tau^*$) over all replicates that alarm before the final step and therefore *mixes* false positives (negative) with correct detections; it is a clean detection-speed measure only for methods whose FP rate is small. ARL_0 is estimated on an independent in-control population and is the unconfounded false-positive metric. M3 (Innovation CUSUM) uses LSTM prediction residuals; no AR model is provided.

Method	FP (pre-τ^*)	TP (correct)	FN (miss)	Mean delay‡	ARL_0
M1: Raw CUSUM	0.603	0.397	0.000	−16.3	119
M2: Whitened CUSUM	0.224	0.772	0.004	+5.9	440
M3: Innovation CUSUM	**0.123**	**0.877**	**0.000**	+10.1	**436**

‡Averaged over any alarm before the final step, including pre-τ^* false alarms; see text.
MSE degradation (M3 model): pre-change 1.002, post-change 2.252, oracle 1.047.

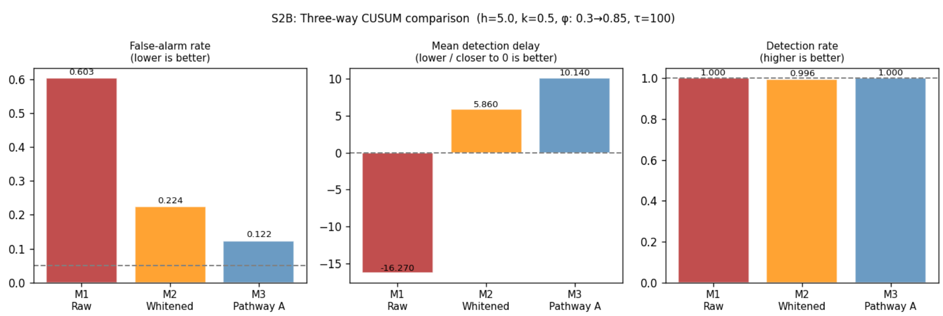

Figure 1. Study CUSUM-CP: three-way CUSUM comparison. Left: false-positive rate (pre-τ^* alarms; dashed line = nominal 5%). Center: mean delay over any alarm (contaminated by false positives for Raw/Whitened; see Table 2). Right: any-alarm rate (*not* correct-detection rate — for Raw this is dominated by pre-τ^* false alarms). The Innovation CUSUM achieves the lowest false-positive rate and, per Table 2, the highest correct-detection rate, without requiring knowledge of the AR model structure.

from $\rho_{\text{before}} = 0.30$ to $\rho_{\text{after}} = 0.85$, while U and the noise remain fixed. The three methods are identical to Section 4.3: M1 (raw Mahalanobis CUSUM), M2 (VAR-whitened CUSUM on OLS residuals), and M3 (Innovation CUSUM: LSTM prediction innovations). For $d > 1$ the scalar CUSUM input is the normalized Mahalanobis score $Z_t = d^{-1}\|\Sigma_0^{-1/2}(X_t - \mu_0)\|^2$, centered and standardized; for M3 the LSTM innovation score is analogously aggregated and calibrated on burn-in in-control sequences. Threshold $h = 5$, $N_{\text{test}} = 1000$ change-point sequences, $N_{\text{ARL}} = 2000$ in-control sequences. ARL_0 is the mean alarm time over in-control sequences of length $5T = 1000$, with alarm-free runs right-censored at the sequence length; censoring biases the estimate downward, so the reported ARL_0 values are conservative.

The deterioration of M1 and M2 with dimension has a clear structural explanation. When the observations are autocorrelated (as under any VAR with $\|\Phi\|_{\text{op}} > 0$), consecutive Mahalanobis scores are positively correlated. The Page CUSUM accumulates this autocorrelation, causing the statistic to drift upward even in-control and shortening ARL_0 well below its nominal value. The problem is compounded in higher dimensions: with d components each contributing correlated mass, the effective autocorrelation of Z_t grows, so ARL_0 collapses as d increases. For M2 (OLS whitening) the same effect applies; moreover, OLS residuals in high-dimensional VAR models carry residual cross-correlation that worsens rather than improves with d. In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are designed to approximate Bayes prediction residuals. By Proposition 2.7, the exact Bayes residuals are MDSs; when the fitted predictor is close to the conditional mean, the fitted residuals are approximate MDSs. Under this mechanism, the CUSUM threshold $h = 5$ is empirically much better calibrated across d in these experiments.

Table 3 and Figure 2 summarize the results, again separating the two error types. At $d = 1$ all three methods are broadly comparable: $\mathrm{ARL}_0 \approx 150$–$200$, false-positive rates of 32%–60%, and correct-detection rates of 40%–68%. As d increases the divergence is dramatic, and it is entirely a false-positive/true-positive story — misses (false negatives) stay below 1% for every method at every d, so nothing is being missed; the question is only whether an alarm, once issued, is correct. At $d = 10$, M1's false-positive rate reaches 99.9%: it alarms before the change on virtually every replicate, so its correct-detection rate collapses to 0.1% ($\mathrm{ARL}_0 = 47$). M2 is worse still — 100% false positives, 0% correct detections ($\mathrm{ARL}_0 = 42$). The Innovation CUSUM moves in the opposite direction: its correct-detection rate *rises* from 67.9% at $d = 1$ to 83.4% at $d = 10$, its false-positive rate stays in the 16%–32% band, and its ARL_0 grows from 187 to 365. The tabulated mean delays reinforce the reading: M3's are small and *positive* (genuine post-τ detections), whereas M1 and M2 report -53 and -58 steps at $d = 10$ — not slow detections but alarms firing almost an entire half-window before the change. The lesson is that the "Detect= 1.00" figure one might report for the raw methods is an artifact of the first-passage rule: an alarm is almost always issued, but for M1/M2 at large d it is almost always a false one. The Innovation CUSUM improves the false-positive side (rising ARL_0, controlled false-positive rate) while *simultaneously* improving the true-positive side, and with misses negligible throughout.

We are careful, however, not to over-read this fixed-h comparison. Two features of the setup favor the innovation detector and are not intrinsic to it: at the shared threshold the detectors operate at very different in-control false-alarm levels (ARL_0 from 42 to 365 at $d = 10$), and the classical detectors estimate an $O(d^2)$ nuisance object (Σ_0, or a VAR matrix) from a short per-sequence in-control window whereas the learned predictor is pretrained on far more data. Study CUSUM-EQARL (Section D.6 of the Supplementary Material) removes both confounds and shows that, at a matched false-alarm level and a matched estimation budget on *these VAR data*, the three detectors are essentially equivalent; the fixed-h collapse is a finite-sample calibration effect, not a structural one. The genuine and provable advantage of learning the predictor appears instead under model misspecification (Proposition 2.9; Study CUSUM-MISSPEC), and in the practically common regime of a short in-control calibration window, where per-sequence estimation of high-dimensional nuisance parameters degrades but a pretrained predictor does not.

4.5. Study ETF: Real Multivariate Application — US Equity Sector Returns

Study ETF addresses the gap that Study CUSUM-DIM leaves open: whether the dimension-stability advantage of the Innovation CUSUM transfers to a real setting where in-control dynamics are non-Gaussian, heteroskedastic, and cross-correlated in a way no parametric VAR fully captures. It should therefore be read as a *stress test* outside the clean theorem assumptions: the sub-Gaussianity of Assumption 3.1(B2) and the conditional-homoskedasticity condition of Corollary 3.11 are known to be violated by heavy-tailed, volatility-clustered equity returns, so the study probes robustness of the qualitative dimension-scaling prediction rather than confirming the bound under its stated hypotheses. Two configurations are examined. In Part A ($d = 1$) we use daily log-returns of the Financial Select

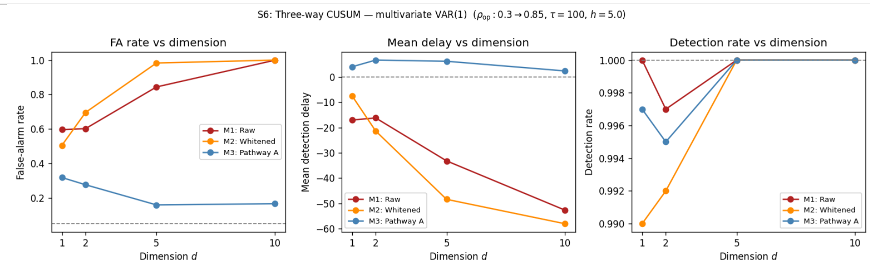

Figure 2. Study CUSUM-DIM: ARL_0 (left) and false-alarm rate (right) as functions of dimension d for the three CUSUM methods. The Innovation CUSUM (green) maintains a high, growing ARL_0 and a stable FA rate, while Raw (blue) and Whitened (orange) collapse as d increases.

Table 3. Study CUSUM-DIM: multivariate CUSUM comparison ($h = 5$, 1000 test sequences, 2000 ARL_0 sequences). The two error types are reported separately: FP = false-positive rate (first alarm before τ); TP = correct-detection rate (first alarm at or after τ); FN = miss rate (no alarm in the window); FP + TP + FN = 1. Delay‡ = mean of (alarm $- \tau$) over any alarm before the final step (negative = pre-τ); it mixes false positives with correct detections and is a clean speed measure only when FP is small. Bold marks the best TP and ARL_0 per block.

d	Method	FP	TP	FN	Delay‡	ARL_0
	Raw (M1)	0.596	0.404	0.000	−17.1	146
1	Whitened (M2)	0.505	0.485	0.010	−7.5	199
	Innovation CUSUM	0.318	**0.679**	0.003	+4.0	**187**
	Raw (M1)	0.602	0.395	0.003	−16.2	145
2	Whitened (M2)	0.696	0.296	0.008	−21.4	126
	Innovation CUSUM	0.276	**0.719**	0.005	+6.6	**231**
	Raw (M1)	0.844	0.156	0.000	−33.2	74
5	Whitened (M2)	0.983	0.017	0.000	−48.4	53
	Innovation CUSUM	0.159	**0.841**	0.000	+6.2	**320**
	Raw (M1)	0.999	0.001	0.000	−52.6	47
10	Whitened (M2)	1.000	0.000	0.000	−58.0	42
	Innovation CUSUM	0.166	**0.834**	0.000	+2.4	**365**

‡Contaminated by pre-τ false alarms for Raw/Whitened; see text.

Sector SPDR (XLF); in Part B ($d = 3$) we add the Technology Select Sector SPDR (XLK) and the Energy Select Sector SPDR (XLE), obtaining a trio with distinct sector exposures.

Daily log-returns are $r_{t,i} = \log(P_{t,i}/P_{t-1,i})$, sourced from Yahoo Finance. The in-control period is 2017-01-01 to 2019-12-31 ($T_0 \approx 754$ trading days), a sustained low-volatility bull-market regime with stable cross-sector correlations. The test window is 2020-01-01 to 2022-06-30 ($T_1 \approx 630$ days). The known change-point is $\tau^* = 32$ steps into the test window, corresponding to 2020-02-19 (the S&P 500 all-time high immediately before the COVID-19 pandemic crash), after which cross-sector correlations rose sharply and the joint return distribution shifted abruptly (Killick, Fearnhead and Eckley, 2012). The COVID break is preferred over the 2008 financial crisis because the Global Financial Crisis (GFC) involved a prolonged pre-crisis deterioration from mid-2007 onward, making it difficult to identify a single clean change-point; the COVID crash was effectively instantaneous at the cross-sector level.

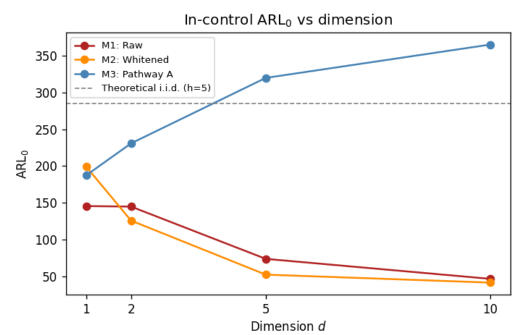

Figure 3. Study CUSUM-DIM: Distribution of detection delay (steps from τ to alarm) across $d \in \{1, 2, 5, 10\}$. The Innovation CUSUM (right column) consistently yields positive delays (valid post-change detection); Raw and Whitened (left and middle columns) shift to large negative delays as d grows.

Because T_0 is large, no bootstrapping is required; the LSTM is trained on rolling windows of length 100 drawn from the in-control period. The threshold and slope parameters are fixed at $h = 5$ and $k = 0.5$, consistent with all other studies. The ARL$_0$ values in Table 4 are estimated from 500 in-control score sequences of length 500, generated by block bootstrap (block length 10) from the pre-COVID calibration scores; alarm-free runs are right-censored at the sequence length, so the estimates are conservative.

The three CUSUM variants from Study CUSUM-DIM are applied in each configuration. M1 (Raw) forms the Mahalanobis score $\mathcal{Z}_t = d^{-1}\|\Sigma_0^{-1/2}(r_t - \mu_0)\|^2 - 1$ using in-control mean μ_0 and Cholesky factor $\Sigma_0^{1/2}$. M2 (Whitened) first applies a VAR(1) whitening step estimated from the in-control period. M3 (Innovation CUSUM) applies the CUSUM to LSTM one-step-ahead prediction residuals standardized by their in-control moments. Part A ($d = 1$) bridges the univariate results (Study CUSUM-CP, and Study NILE of the Supplementary Material) and the multivariate synthetic results (Study CUSUM-DIM); Part B ($d = 3$) tests whether the advantage is maintained under genuine cross-sectional dependence.

Table 4 summarizes the results across both configurations ($h = 5$, $k = 0.5$). In Part A ($d = 1$), the Innovation CUSUM alarms at $t = 17$, fifteen steps before the officially designated COVID peak $\tau^* = 32$. Under the formal definition of Section 3.1, any alarm before τ^* constitutes a false alarm. The negative delay should nonetheless be interpreted cautiously: the S&P 500 sell-off began in early February, two weeks before the officially designated February 19 peak, so the true onset of the distributional shift is itself uncertain and the pre-τ^* alarm may reflect genuine early-warning signals in the LSTM innovation process. This illustrates the practical difficulty of separating a false positive from an early true positive when the change onset is itself ill-defined: whether one treats the $t = 17$ alarm as a false alarm or a timely detection depends on the cost asymmetry of the application — in a risk-monitoring setting a two-week-early warning of a market dislocation is typically the desirable error to make, whereas a late alarm (a false negative at the peak) is not. M1 and M2 alarm at $t = 36$ (+4 delay), after the peak. The cleaner, onset-independent comparison is the in-control ARL$_0$, estimated entirely within the pre-COVID regime and therefore a pure false-positive-rate measure: M1 and M2 attain 38 and 36 versus the Innovation CUSUM's 43, indicating more frequent false alarms.

In Part B ($d = 3$), the pattern is amplified. M1 and M2 alarm at $t = 24$ (delay -8) while the Innovation CUSUM again alarms at $t = 17$ (delay -15), earlier because the cross-sectional energy (XLE) signal accelerates the Mahalanobis score under the joint distribution shift. Crucially, the Innovation CUSUM's ARL$_0$ rises from 43 to 67 as d goes from 1 to 3, whereas M1 rises only from 38 to 58 and M2 from 36 to 49. The gap in ARL$_0$ between the Innovation CUSUM and M1/M2 widens with dimension, consistent

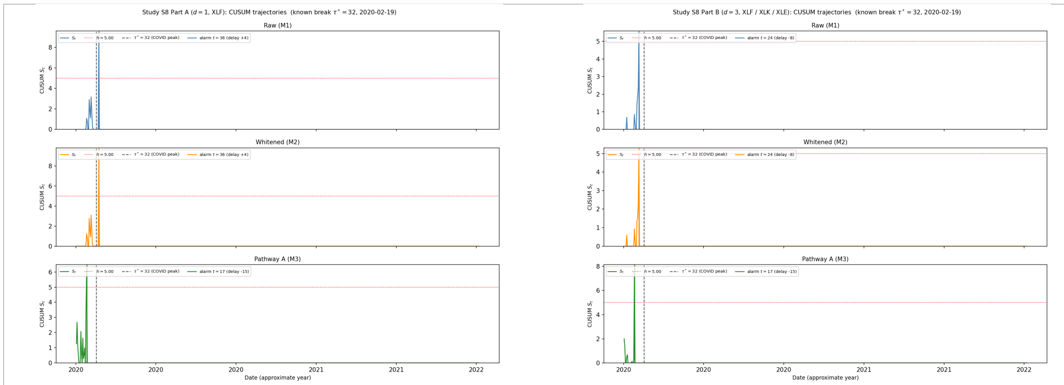

Figure 4. Study ETF: CUSUM trajectories on the COVID test window (2020-01-01 to 2022-06-30); the vertical dashed line marks $\tau^* = 32$ (2020-02-19). *Left* (Part A, $d = 1$, XLF): the Innovation CUSUM (bottom) crosses threshold $h = 5$ at $t = 17$, fifteen steps before the official market peak. *Right* (Part B, $d = 3$, XLF/XLK/XLE): the Innovation CUSUM again alarms at $t = 17$, and its ARL_0 gap over M1/M2 widens relative to Part A, consistent with the dimension-stability prediction of Corollary 3.11.

with Corollary 3.11 and the simulation evidence in Table 3. Figure 4 displays the CUSUM trajectories for both configurations: in each panel the rapid rise of the Innovation CUSUM statistic in late January–early February 2020 is visible, whereas M1 and M2 remain flat until after the February 19 peak.

Table 4. Study ETF: US equity sector ETF CUSUM results ($h = 5$, $k = 0.5$). Known change-point $\tau^* = 32$ (2020-02-19). ARL_0 estimated from 500 block-bootstrapped in-control score sequences of length 500, right-censored at the sequence length. Delay = alarm time − τ^* (negative = before the break).

Config	Method	Alarm t	Delay	ARL_0
Part A ($d = 1$, XLF)	Raw (M1)	36	+4	38.1
	Whitened (M2)	36	+4	36.0
	Innovation CUSUM (M3)	**17**	−15	**43.5**
Part B ($d = 3$, XLF/XLK/XLE)	Raw (M1)	24	−8	58.1
	Whitened (M2)	24	−8	49.3
	Innovation CUSUM (M3)	**17**	−15	**67.5**

4.6. Summary and Interpretation

Taken together, the twelve studies — four here, eight in the Supplementary Material — give a finite-sample picture of how the hidden-state innovation, the predictive-decay profile J_t, and the Innovation CUSUM behave across synthetic and real settings ranging from univariate AR to non-Gaussian high-dimensional equity returns. The synthetic studies draw on many independent replications and support statistical comparison; Study ETF and Study NILE (Section D.4 of the Supplementary Material), by contrast, each examine a single real change-point event and are illustrative case studies — evidence that the pipeline runs end-to-end on real data, not that the method has a typical real-world advantage, which would require many more real change events than either series provides.

The dimensional scaling documented at fixed h in Study CUSUM-DIM is striking — raw and whitened ARL_0 collapse as d grows while the Innovation CUSUM's rises — but it must be read with care. As Study CUSUM-EQARL (Section D.6 of the Supplementary Material) shows, at a matched false-alarm level and a matched estimation budget the three detectors are essentially equivalent on these VAR data: the fixed-h collapse reflects the classical detectors operating at a far less conservative point and estimating high-dimensional nuisance parameters from a short in-control window, not a structural deficiency. Corollary 3.11's serial-dependence bias is real, but once the reference level is calibrated and the nuisance parameters are well estimated it does not, in this regime, manifest as a dimension problem. The advantage of learning the predictor that *does* survive a fully fair comparison is twofold: under nonlinearity in the conditional mean, a fixed linear residualization carries an irreducible innovation defect while the learned predictor does not (Proposition 2.9, borne out by the whiteness ordering of Study CUSUM-MISSPEC); and in the practically common short-window regime, the learned predictor sidesteps the ill-conditioned per-sequence estimation that degrades the classical scores. The in-control guarantee is, moreover, sharp: the exponent $2(k-\eta)/\sigma^2$ of Theorem 3.2 is the Lundberg coefficient and coincides with the exact exponential rate of the classical i.i.d. CUSUM (Siegmund, 1985, Ch. 3), obtained here by a renewal argument that leaves no looseness in the exponent.

Study KAL (Section D.3 of the Supplementary Material) explains why high persistence is structurally difficult: the analytical decay profile J_t decreases slowly, so hidden-state proxies must integrate information over a long history; the LSTM outperforms the EM-Kalman estimator at moderate persistence but the Kalman recovers its advantage when $\phi \approx 0.95$. Study J-Surr (Section D.9 of the Supplementary Material) shows that KLIEP-type corrections achieve NMSE < 1 under smooth non-stationarity but deteriorate under abrupt change, delimiting where density-ratio reweighting can substitute for retraining.

5. Conclusion

Two practical gaps motivated this work. First, recurrent networks are routinely used for sequential prediction, yet prior work on their stability and expressivity supplies no martingale-innovation account of the recurrent state usable for calibrating downstream inference under explicit residual conditions — rather than a full account of what the hidden state encodes in general. Second, classical change-point detectors are calibrated for independent or fully specified parametric input, so their false-alarm guarantees break down under unknown serial dependence, the more so as the observation dimension grows.

To address the first gap, Theorem 2.3 establishes the Doob decomposition $h_t = D_t + M_t$ with an explicit geometric L^2 innovation bound and stationary covariance, and Corollary 2.6 the functional central limit theorem for accumulated innovations. To address the second, Theorems 3.2 and 3.6 give a sharp Lundberg-rate in-control ARL lower bound and a matching detection-delay upper bound, combining (Corollary 3.7) into the exponential-false-alarm / logarithmic-delay trade-off for a detector with no parametric model for the observation dynamics; Theorem 3.9 shows this trade-off is order-optimal in Lorden's minimax framework, and exactly minimax optimal under a Gaussian innovation shift. Propositions 2.8 and 2.9 link state stabilization to residual drift and identify when a learned predictor attains a larger guaranteed drift margin — and hence a larger guaranteed ARL exponent — than a fixed linear filter, and Corollary 3.11 treats the multivariate score. Empirical studies — synthetic autoregressive and vector-autoregressive processes, a nonlinear misspecified process, the Nile discharge series, and US equity sector returns around the 2020 COVID crash — illustrate these mechanisms.

What appears to be new is the *combination* of these ingredients: to our knowledge, no prior work joins recurrent-state innovation theory with sequential CUSUM calibration to yield a model-free monitoring guarantee for a detector built on a learned predictor.

Several questions remain open. The dimension-scaling advantage at fixed threshold is, under a fully fair comparison, a finite-sample calibration effect rather than a structural one (Study CUSUM-EQARL);

characterizing when short-window nuisance estimation makes the learned score preferable is left open, as is a fully non-stationary theory beyond the locally stationary approximation of Corollary 2.15. The in-control exponent is already sharp (Theorem 3.2); the remaining slack is in the delay-bound constant, which the overshoot distribution of the reflected random walk could sharpen, and relaxing the sub-Gaussianity and conditional-homoskedasticity conditions of Corollary 3.11 to heavy-tailed or heteroskedastic innovations is a natural next step with direct relevance to finance.

More broadly, what a learned sequential state encodes in probabilistic terms is not specific to RNNs: it arises for transformer attention heads, diffusion time-series latents, and language-model context representations. A martingale characterization extending to those state representations would place inference and monitoring on such models on the same rigorous footing.

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